Quantum Scientific Computing (QSC2026)

Homework 1 — Solutions

These solutions show the intended level of detail; many admit equivalent routes. Numerical answers are exact where a closed form exists and rounded to four digits otherwise.

Problem 1 (Single qubit). $|\psi\rangle = a\,|0\rangle + b\,|1\rangle$ with $a = \cos\frac{\pi}{8}$, $b = \mathrm{e}^{\mathrm{i}\pi/4}\sin\frac{\pi}{8}$.

Solution. **(a)** $|a|^2 + |b|^2 = \cos^2\frac{\pi}{8} + \sin^2\frac{\pi}{8} = 1$. The computational-basis probabilities are $\mathbb{P}(0) = \cos^2\frac{\pi}{8} = \frac{2+\sqrt{2}}{4} \approx 0.8536$ and $\mathbb{P}(1) = \sin^2\frac{\pi}{8} = \frac{2-\sqrt{2}}{4} \approx 0.1464$.

(b) With $\bar{a}b = \cos\frac{\pi}{8}\sin\frac{\pi}{8}\,\mathrm{e}^{\mathrm{i}\pi/4} = \frac{1}{2}\sin\frac{\pi}{4}\,\mathrm{e}^{\mathrm{i}\pi/4} = \frac{\sqrt{2}}{4}\mathrm{e}^{\mathrm{i}\pi/4}$,

$$\langle X\rangle = 2\Re(\bar{a}b) = \tfrac{\sqrt{2}}{4}\cdot 2\cos\tfrac{\pi}{4} = \tfrac{1}{2}, \quad \langle Y\rangle = 2\Im(\bar{a}b) = \tfrac{1}{2}, \quad \langle Z\rangle = |a|^2 - |b|^2 = \cos\tfrac{\pi}{4} = \tfrac{\sqrt{2}}{2}.$$

Bloch vector $\mathbf{r} = (\frac{1}{2}, \frac{1}{2}, \frac{\sqrt{2}}{2})$, with $|\mathbf{r}|^2 = \frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$ (pure state).

(c) $\mathbb{P}(+) = |\langle +|\psi\rangle|^2 = \frac{1}{2}\,|a+b|^2 = \frac{1}{2}(1 + 2\Re(\bar{a}b)) = \frac{1}{2}(1 + \langle X\rangle) = \frac{1}{2}(1+\frac{1}{2}) = \frac{3}{4}$. (The identity $\mathbb{P}(+) = \frac{1}{2}(1+\langle X\rangle)$ is the general rule for an X-basis measurement.)

(d) Model one Z-measurement as a random variable $m \in \{+1, -1\}$ ($+1$ for outcome 0, -1 for outcome 1), so $\mathbb{E}[m] = \mathbb{P}(0) - \mathbb{P}(1) = \langle Z\rangle$ and, since $m^2 = 1$ always, $\mathrm{Var}(m) = \mathbb{E}[m^2] - \mathbb{E}[m]^2 = 1 - \langle Z\rangle^2 = 1 - \frac{1}{2} = \frac{1}{2}$. Averaging N independent shots, the estimator $\hat{Z} = \frac{1}{N}\sum_{i=1}^{N} m_i$ is unbiased with standard error $\sqrt{\mathrm{Var}(m)/N} = \sqrt{(1-\langle Z\rangle^2)/N}$. By the central limit theorem $\hat{Z}$ lies within $\pm\epsilon$ of $\langle Z\rangle$ at a confidence level set by z (e.g. $z \approx 2$ for $\approx 95\%$) once $z\sqrt{\mathrm{Var}(m)/N} \le \epsilon$, i.e.

$$N \;\ge\; \frac{z^2\,(1-\langle Z\rangle^2)}{\epsilon^2} \;=\; \frac{4\cdot\frac{1}{2}}{(0.05)^2} \;=\; 800.$$

(A distribution-free Hoeffding bound gives the same order, $N \ge \ln(2/\delta)/(2\epsilon^2)$, and the worst case $\mathrm{Var}(m) \le 1$ needs at most $N \le z^2/\epsilon^2 = 1600$.) The essential point is the *shot-noise* (standard quantum limit) law $N = \Theta(1/\epsilon^2)$: halving the target error *quadruples* the number of measurements—exactly the scaling that amplitude estimation (Problem 10) improves to $\Theta(1/\epsilon)$.

Problem 2 (Separability test). A state $a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle$ factorizes iff $ad = bc$.

Solution. **(a)** $a = b = c = d = \frac{1}{2}$: $ad = \frac{1}{4} = bc$, so it is a *product* state, $= \frac{1}{2}(|0\rangle + |1\rangle)(|0\rangle + |1\rangle) = |+\rangle \otimes |+\rangle$.

(b) $a = \frac{1}{\sqrt{2}}$, $d = -\frac{1}{\sqrt{2}}$, $b = c = 0$: $ad = -\frac{1}{2} \neq 0 = bc$, so it is *entangled* (the Bell state $|\Phi^-\rangle$).

(c) $a = b = c = \frac{1}{2}$, $d = -\frac{1}{2}$: $ad = -\frac{1}{4} \neq \frac{1}{4} = bc$, so it is *entangled*.

(d) If $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = (\alpha\,|0\rangle + \beta\,|1\rangle)(\gamma\,|0\rangle + \delta\,|1\rangle)$ then matching coefficients gives $\alpha\gamma = \beta\delta = \frac{1}{\sqrt{2}}$ (so $\alpha, \beta, \gamma, \delta \neq 0$) but $\alpha\delta = \beta\gamma = 0$ (coefficients of $|01\rangle, |10\rangle$), a contradiction. Hence no factorization exists; equivalently $ad - bc = \frac{1}{2} \neq 0$.

Problem 3 (Gates, reversibility, no-cloning).

Solution. **(a)** $(H \otimes \mathbb{I})\,|00\rangle = |+\rangle\,|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$; then CNOT (control first) sends $|10\rangle \mapsto |11\rangle$, giving $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle$.

(b) Toffoli: $|a,b,c\rangle \mapsto |a,b,\, c\oplus(a\wedge b)\rangle$, so with $c=0$ it writes $a\wedge b$ onto the third qubit. It is its own inverse, hence reversible. The bare map $(a,b)\mapsto a\wedge b$ is $2\to 1$ and not injective (both 01 and 00 give 0), so it cannot be a unitary on two qubits; keeping a,b and writing the result on a fresh qubit restores invertibility.

(c) Take $U=\mathrm{CNOT}$ and $|s\rangle=|0\rangle$: then $U|0\rangle|0\rangle=|00\rangle$ and $U|1\rangle|0\rangle=|11\rangle$ as required. On a general input, $U(\alpha|0\rangle+\beta|1\rangle)|0\rangle=\alpha|00\rangle+\beta|11\rangle$, whereas a true copy would be $(\alpha|0\rangle+\beta|1\rangle)^{\otimes 2}=\alpha^2|00\rangle+\alpha\beta|01\rangle+\alpha\beta|10\rangle+\beta^2|11\rangle$. These agree only when $\alpha\beta=0$, i.e. for basis inputs; U entangles instead of copying. Linearity forbids a universal cloner—this is the no-cloning theorem.

Problem 4 (Teleportation). Message $|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$ on qubit 1; shared $\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle)$ on qubits 2 (Alice) and 3 (Bob).

Solution. **(a)** The joint state $\frac{1}{\sqrt{2}}(\alpha|000\rangle+\alpha|011\rangle+\beta|100\rangle+\beta|111\rangle)$ re-expressed in the Bell basis of qubits $1,2$ is

$$\tfrac{1}{2}\Big[|\Phi^+\rangle(\alpha|0\rangle+\beta|1\rangle)+|\Phi^-\rangle(\alpha|0\rangle-\beta|1\rangle)+|\Psi^+\rangle(\alpha|1\rangle+\beta|0\rangle)+|\Psi^-\rangle(\alpha|1\rangle-\beta|0\rangle)\Big]_3.$$

With the standard $\mathrm{CNOT}_{1\to 2}$ then H_1 measurement giving bits (m_1,m_2), the four outcomes and Bob's pre-correction states are $(0,0)\to|\psi\rangle$, $(0,1)\to X|\psi\rangle$, $(1,0)\to Z|\psi\rangle$, $(1,1)\to XZ|\psi\rangle$, each with probability $\frac{1}{4}$.

(b) For $(m_1,m_2)=(1,1)$ the pre-correction state is $XZ|\psi\rangle=\alpha|1\rangle-\beta|0\rangle$. Bob applies X then Z (i.e. $Z^{m_1}X^{m_2}$): $X(\alpha|1\rangle-\beta|0\rangle)=\alpha|0\rangle-\beta|1\rangle=Z|\psi\rangle$, then $Z(\alpha|0\rangle-\beta|1\rangle)=\alpha|0\rangle+\beta|1\rangle=|\psi\rangle$. Restored.

(c) Before the bits arrive, all four outcomes are equally likely, so Bob's density operator is $\frac{1}{4}\sum_{P\in\{\mathbb{I},X,Z,XZ\}}P|\psi\rangle\langle\psi|P^\dagger=\frac{1}{2}\mathbb{I}$, the maximally mixed state—independent of $|\psi\rangle$. Nothing Alice does changes Bob's local statistics until the two classical bits (sent no faster than light) arrive; hence no faster-than-light signaling.

Problem 5 (Phase kickback; Deutsch; Deutsch–Jozsa).

Solution. **(a)** $U_f|x\rangle\frac{1}{\sqrt{2}}(|0\rangle-|1\rangle)=\frac{1}{\sqrt{2}}|x\rangle(|f(x)\rangle-|1\oplus f(x)\rangle)=(-1)^{f(x)}|x\rangle\frac{1}{\sqrt{2}}(|0\rangle-|1\rangle)=(-1)^{f(x)}|x\rangle|-\rangle$, since the pair $(|f(x)\rangle,|1\oplus f(x)\rangle)$ equals $(|0\rangle,|1\rangle)$ if $f(x)=0$ and $(|1\rangle,|0\rangle)$ if $f(x)=1$.

(b) The four functions: $f\equiv 0$ and $f\equiv 1$ are *constant*; $f(x)=x$ and $f(x)=1-x$ are *balanced*. Deutsch prepares $|+\rangle|-\rangle$; phase kickback gives the first qubit $\propto(-1)^{f(0)}|0\rangle+(-1)^{f(1)}|1\rangle$. Applying H: if $f(0)=f(1)$ the qubit is $\pm|+\rangle\xrightarrow{H}\pm|0\rangle$ (measure 0); if $f(0)\neq f(1)$ it is $\pm|-\rangle\xrightarrow{H}\pm|1\rangle$ (measure 1). So the single query returns 0 iff f is constant.

(c) In Deutsch–Jozsa the amplitude of $|0^n\rangle$ is $2^{-n}\sum_x(-1)^{f(x)}$: it equals ±1 (probability 1) when f is constant and 0 when f is balanced. The promise "f is constant or balanced" is what makes the one-query, all-zeros test exact.

Problem 6 (Grover search, $N=16$).

Solution. **(a)** $\sin\frac{\theta}{2}=\frac{1}{\sqrt{16}}=\frac{1}{4}$, so $\theta=2\arcsin\frac{1}{4}\approx 0.5054$ ($\approx 28.96^\circ$). $R_{x_0}=\mathbb{I}-2|x_0\rangle\langle x_0|$ reflects about the axis $|\psi_0^\perp\rangle$ and $R_{\psi_0}=2|\psi_0\rangle\langle\psi_0|-\mathbb{I}$ reflects about $|\psi_0\rangle$; a product of two reflections is a rotation by twice the angle between their mirror axes. Since $|\psi_0\rangle$ sits at angle $\theta/2$ from $|\psi_0^\perp\rangle$, $G=R_{\psi_0}R_{x_0}$ rotates by θ toward $|x_0\rangle$.

(b) $p_k=\sin^2((2k+1)\theta/2)$:

$$p_0=0.0625,\ p_1=0.4727,\ p_2=0.9084,\ p_3=0.9613,\ p_4=0.5817,\ p_5=0.1255.$$

The optimum is $k^* = 3$ ($p_3 \approx 0.961$). At $k = 4, 5$ the angle $(2k+1)\theta/2$ passes $\frac{\pi}{2}$, so the state rotates *past* $|x_0\rangle$ and the success probability falls back (over-rotation).

(c) For $N = 2^n$ with one marked item, $\theta/2 = \arcsin(1/\sqrt{N}) \approx 1/\sqrt{N}$; maximizing wants $(2k+1)\theta/2 \approx \frac{\pi}{2}$, giving $k^* \approx \frac{\pi}{4}\sqrt{N}$ and query complexity $\mathcal{O}(\sqrt{N})$.

Problem 7 (Hadamard test).

Solution. **(a)** After H on the ancilla and controlled-U the state is $\frac{1}{\sqrt{2}}(|0\rangle|\psi\rangle + |1\rangle U|\psi\rangle)$; a second H gives $\frac{1}{2}\big[|0\rangle(|\psi\rangle + U|\psi\rangle) + |1\rangle(|\psi\rangle - U|\psi\rangle)\big]$, so $p(0) = \frac{1}{4}\||\psi\rangle + U|\psi\rangle\|^2 = \frac{1}{4}(2 + 2\Re\langle\psi|U|\psi\rangle) = \frac{1}{2}(1 + \Re\langle\psi|U|\psi\rangle)$. Inserting $S^\dagger = \mathrm{diag}(1, -\mathrm{i})$ on the ancilla between the two Hadamards replaces $\Re$ by $\Im$: $p(0) = \frac{1}{2}(1 + \Im\langle\psi|U|\psi\rangle)$.

(b) $U = S = \mathrm{diag}(1, \mathrm{i})$, $|\psi\rangle = |+\rangle$: $\langle+|S|+\rangle = \frac{1}{2}(\langle 0| + \langle 1|)(|0\rangle + \mathrm{i}|1\rangle) = \frac{1}{2}(1+\mathrm{i})$, so $\Re = \Im = \frac{1}{2}$. The real-part test predicts $p(0) = \frac{1}{2}(1 + \frac{1}{2}) = \frac{3}{4}$ and the imaginary-part variant also $\frac{3}{4}$, consistent with $\langle+|S|+\rangle = (1+\mathrm{i})/2$.

Problem 8 (QFT and phase estimation).

Solution. **(a)** For $n = 3$, $\mathrm{QFT}|j\rangle = \frac{1}{\sqrt{8}}(|0\rangle + \mathrm{e}^{2\pi\mathrm{i}\,0.j_3}|1\rangle)(|0\rangle + \mathrm{e}^{2\pi\mathrm{i}\,0.j_2 j_3}|1\rangle)(|0\rangle + \mathrm{e}^{2\pi\mathrm{i}\,0.j_1 j_2 j_3}|1\rangle)$. The circuit uses 3 Hadamards and $\binom{3}{2} = 3$ controlled-phase gates (R_2, R_3), plus the final bit-reversal swap.

(b) $\varphi = \frac{3}{8}$, $t = 3$. After the controlled powers the ancilla is $\frac{1}{\sqrt{8}}\sum_{j=0}^{7}\mathrm{e}^{2\pi\mathrm{i}(3/8)j}|j\rangle = \mathrm{QFT}|3\rangle$, because $\varphi = k/8$ with $k = 3$. Applying QFT^{-1} returns $|3\rangle = |011\rangle$, measured with probability 1. The readout is *exact* because $\varphi = k/2^t$ is a dyadic rational with a t-bit expansion.

(c) Using $t = n + \lceil\log_2(2 + \frac{1}{2\delta})\rceil$ for success $\geq 1 - \delta$: with $n = 6$, $\delta = 0.1$ we get $\lceil\log_2(2+5)\rceil = \lceil\log_2 7\rceil = 3$, hence $t = 9$ ancilla qubits.

Problem 9 (Ground-state energy estimation).

Solution. **(a)** $U|E_j\rangle = \mathrm{e}^{-\mathrm{i}Ht}|E_j\rangle = \mathrm{e}^{-\mathrm{i}E_j t}|E_j\rangle = \mathrm{e}^{2\pi\mathrm{i}\varphi_j}|E_j\rangle$ with $\varphi_j = -\frac{E_j t}{2\pi} \bmod 1$. QPE estimates φ_j; inverting, $E_j = -\frac{2\pi}{t}\varphi_j$ (taken in the window chosen so $E_j t/2\pi \in [0, 1)$).

(b) By linearity QPE on $|\phi\rangle = \sum_j c_j|E_j\rangle$ collapses onto $|E_j\rangle$ (returning φ_j) with probability $|c_j|^2$; the ground phase φ_0 thus appears with probability $p_0 = |c_0|^2 = 0.4$ per run. For R independent runs, $\mathbb{P}(\text{seen at least once}) = 1 - (1-p_0)^R \geq 0.99$ requires $(0.6)^R \leq 0.01$, i.e. $R \geq \ln 0.01/\ln 0.6 \approx 9.02$, so $R = 10$ runs suffice.

(c) Energy estimation only reads *some* eigenvalue each run and keeps the smallest observed; it needs the ground state to have nonnegligible overlap, but *not* a spectral gap, since one is not trying to resolve neighboring levels. Preparing the ground state (filtering out the excited components) does need a gap to separate E_0 from the rest.

Problem 10 (Amplitude estimation).

Solution. **(a)** Write $A|0^n\rangle = \sin\frac{\theta}{2}|\phi_{\text{good}}\rangle + \cos\frac{\theta}{2}|\phi_{\text{bad}}\rangle$, so $P = \sin^2\frac{\theta}{2}$. On the plane $\mathrm{span}\{|\phi_{\text{good}}\rangle, |\phi_{\text{bad}}\rangle\}$ the iterate $G = R_{\phi_0}R_{\text{good}}$ is a rotation by θ with eigenvalues $\mathrm{e}^{\pm\mathrm{i}\theta}$; the prepared state is an equal superposition of the two eigenvectors, so QPE on G returns $\pm\theta$ and hence $P = \sin^2(\theta/2)$.

(b) For $\epsilon = 10^{-3}$: amplitude estimation uses $\mathcal{O}(1/\epsilon) \approx 10^3$ applications of A (and of G), whereas classical Monte Carlo needs $\mathcal{O}(1/\epsilon^2) \approx 10^6$ samples—a factor $\approx 10^3$ fewer queries.

(c) The Hadamard test encodes $\Re\langle\psi|U|\psi\rangle$ in the bias $p(0) = \frac{1}{2}(1 + \Re\langle\psi|U|\psi\rangle)$. Estimating $p(0)$ by repetition costs $\mathcal{O}(1/\epsilon^2)$ runs (shot noise); running amplitude estimation on the very same circuit estimates $p(0)$—hence $\Re\langle\psi|U|\psi\rangle$—to precision ϵ with $\mathcal{O}(1/\epsilon)$ coherent applications. This is the quadratic (Heisenberg-limited) improvement over plain repetition.